Dongjie Yu (余冬杰)

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Education

The University of Hong Kong (HKU) Sep. 2023 – Present
Ph.D. Student in Computer Science *Advisor: Prof. Jia Pan*

Tsinghua University (THU) Sep. 2020 – Jun. 2023
Master of Science in Mechanical Engineering *Advisor: Prof. Shengbo Li & Prof.Jianyu Chen*

Tsinghua University Sep. 2016 – Jul. 2020
Bachelor of Engineering in Vehicle Engineering *Advisor: Prof. Shengbo Li*

- GPA:3.75/4, rank: 4/65 **(top 10%)**
- Awarded with the Excellent Graduate of Tsinghua University **(top 10%)**

Tsinghua University Sep. 2017 – Jul. 2020
Minor in Computer Application Technology

Research Interest

Robotic Manipulation, Imitation Learning (IL), Reinforcement Learning (RL), Safe RL, Learning-based Control

Honors and Awards

Hong Kong PhD Fellowship Scheme (HKPFS), RGC of Hong Kong	2023-2027
HKU Presidential PhD Scholarship (HKU-PS), HKU	2023-2027
Y S and Christabel Lung Postgraduate Scholarship, HKU	2023
Excellent Graduate, Tsinghua University **(top 10%)**	2020
Outstanding Bachelor Thesis, School of Vehicle and Mobility **(top 20%)**	2020
Scholarship of Academic Excellence, Tsinghua University **(top 30%)**	2017, 2018, 2019
Scholarship of Freshmen, School of Vehicle and Mobility **(2/65)**	2016

Research and Project Experiences

RL Post-training for Robot Manipulation Policies (RL-100, ZPRL) Jun. 2025 – Now
Intern, Shanghai Qizhi Institute (SQZ)

- Verify (collecting demo, iterative offline, online) RL-100 (RL on 3D diffusion policies) on a two-stage orange juicing task, consecutively serving customers for 7 hours in a shopping mall.
- Develop an RL algorithm steering base imitation policies through a bottleneck latent interface (ZPRL), with higher asymptotic returns, learning efficiency and smooth exploration.
- Real-world ZPRL on three challenging tasks: placing orange, dynamic flipping egg, and bimanual opening box.

Fine-grained Industrial Bimanual Manipulation (BiKC) Oct. 2023 – Jan. 2025
Postgraduate Researcher, School of Computing and Data Science, HKU

- Propose a hierarchical imitation learning framework for bimanual manipulation: a high-level keypose predictor forecasting next bimanual keypose to guide low-level trajectory generation and improve per-stage reliability.
- Formulate the low-level policy as a consistency model to avoid iterative denoising in diffusion policy and enhance per-step efficiency during execution. (Paper 5)
- Simulation and real-world experiments on ALOHA indicate that the proposed framework improves sub-stage and overall success rates while maintaining real-time reaction. Also verify the framework on a dual franka-sigma system.

Safe Model-based RL with a Distributional Reachability Certificate (DRPO) May. 2022 – Jul. 2023
Postgraduate Researcher, Intelligent Driving Lab, School of Vehicle and Mobility, THU

- Extend the reachability certificate proposed in Project RCRL to a distributional setting to realize robust safety against the model uncertainty in model-based RL.
- Propose a safe RL framework to resolve distributional reachability constraints and leverage a shield policy derived from the certificate.
- Evaluate DRPO on several safe RL benchmarks and show that training time violations are reduced significantly while cumulative rewards are maintained. (Paper 3)

Reachability Constrained RL for Safety-critical Dynamical Systems (RCRL) Nov. 2021 – May. 2022

Postgraduate Researcher, Intelligent Driving Lab, School of Vehicle and Mobility, THU

- Introduce reachability certificate from Hamilton-Jacobi Reachability to describe forward-invariant state subspace in constrained RL and guide policy updates.
- Evaluate RCRL on 2D quadrotor tracking tasks and robots navigation tasks in simulation. RCRL converges to zero-violation policies with competitive performance. (Paper 6)

Integrated Decision and Control for Autonomous Driving (IDC) Jan. 2021 – Dec. 2021

Postgraduate Researcher, Intelligent Driving Lab, School of Vehicle and Mobility, THU

- Design a static path planner for general intersections, together with a velocity-choosing mechanism addressing switch among traffic lights. Results serve as tracking reference for MPC or RL-trained policies. (Patent CN202110990214.8)
- Implemente an attention-based model as decision-making neural networks backbone to deal with dynamic number of surrounding traffic participants. (Paper 7)

Permutation Invariant State Representation for Autonomous Driving (ESC) Jun. 2020 – Dec. 2020

Postgraduate Researcher, Intelligent Driving Lab, School of Vehicle and Mobility, THU

- Propose a permutation-invariant representation method (called ESC) to encode dynamic number of participants in autonomous driving. Prove injectivity and representation capability of ESC with theory and empirical experiments. Compared to fixed-permutation methods, ESC reduces approximation error by 62.2%. (Paper 4)
- Adopt attention-based mechanism to address equal weighting in ESC. (Applied in Project IDC)

Publications

Journal Papers (* equal contribution)

1. Lei K., Li H., **Yu D.**, et al. RL-100: Performant Robotic Manipulation with Real-World Reinforcement Learning (2026). Under review.
2. Xu H., Chen Y., **Yu D.**, et al. BiKC+: Bimanual Hierarchical Imitation with Keypose-Conditioned Coordination-Aware Consistency Policies (2025). IEEE Transactions on Automation Science and Engineering.
3. **Yu D.**, Zou W., Yang Y., et al. Safe Model-based Reinforcement Learning with an Uncertainty-Aware Reachability Certificate (2023). IEEE Transactions on Automation Science and Engineering (T-ASE).
4. Duan J.*, **Yu D.***, Li S. E., et al. Fixed-Dimensional and Permutation Invariant State Representation of Autonomous Driving (2021). IEEE Transactions on Intelligent Transportation Systems (T-ITS).

Conference & Workshop Papers

5. **Yu D.***. Xu H.*, Chen Y., et al. BiKC: Keypose-Conditioned Consistency Policy for Bimanual Robotic Manipulation (2024). International Workshop on the Algorithmic Foundations of Robotics (WAFR).
6. **Yu D.***, Ma H.*, Li S. E., et al. Reachability Constrained Reinforcement Learning (2022). International Conference on Machine Learning (ICML).
7. Jiang J., Ren Y., Guan Y., Li S. E., Yin Y., **Yu D.**, et al. Integrated Decision and Control at Multi-Lane Intersections with Mixed Traffic Flow (2021). International Conference on Intelligent Vehicles (ICoIV). **(Best student paper award)**

Book Chapter

8. Chapter 9.6.2 (5 pages about application of HJ Reachability in Constrained RL) and Chapter 11.7 (6 pages about common RL libraries and benchmarks) of *Reinforcement Learning for Sequential Decision and Optimal Control* by Shengbo Eben Li. Springer, 2023.

Skills

Programming Language: Python, C/C++, MATLAB
Software and Platform: Linux, PyTorch, MATLAB & Simulink, Git
English: IELTS (8.0 (6.5))
Hobby: Football

Service

Reviewer, 6 manuscripts

- Journal: T-RO (1), T-II (1)
- Conference: ICRA (3), IROS (1)

Member, Tech Service Team of Work-Study Program at THU (2020 – 2023)

- Excellent Team Member of the Month ×3, Excellent Team Member of the Semester ×2
- **Outstanding Individual** of Work-Study Program in 2021 (top 10%).